

Database System Engineering and Distributed Backend Development Course

Code: 25CS1302E

ROLL NO : 2520030255

NAME : V SAI NIKHIL

WEEK-4 MYSQL VIEWS

```
1 • CREATE DATABASE IF NOT EXISTS employeedb1;
2 • USE employeedb1;
3
4 • CREATE TABLE IF NOT EXISTS departments (
5     dept_id INT PRIMARY KEY,
6     dept_name VARCHAR(50)
7 );
8
9 • CREATE TABLE IF NOT EXISTS employees (
10     emp_id INT PRIMARY KEY,
11     emp_name VARCHAR(50),
12     dept_id INT,
13     salary DECIMAL(10,2)
14 );
15
16 • INSERT IGNORE INTO departments VALUES
17     (1,'CSE'),
18     (2,'ECE'),
19     (3,'EEE');
20
21 • INSERT IGNORE INTO employees VALUES
22     (101,'Rahul',1,50000),
23     (102,'Priya',2,65000),
24     (103,'Anil',1,55000),
25     (104,'Sneha',3,70000),
```

Output

Action Output

#	Time	Action	Message
✓ 96	13:12:20	CREATE DATABASE IF NOT EXISTS employeedb1	1 row(s) affected
✓ 97	13:12:20	USE employeedb1	0 row(s) affected
✓ 98	13:12:20	CREATE TABLE IF NOT EXISTS departments (dept_id INT PRIMARY KEY, dept_name VARCHAR(50))	0 row(s) affected
✓ 99	13:12:20	CREATE TABLE IF NOT EXISTS employees (emp_id INT PRIMARY KEY, emp_name VARCHAR(50), d...	0 row(s) affected
✓ 100	13:12:20	INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,'EEE')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 101	13:12:20	INSERT IGNORE INTO employees VALUES (101,'Rahul',1,50000), (102,'Priya',2,65000), (103,'Anil',1,55000), (1...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

```
1 SHOW TABLES;
2 • SELECT * FROM departments;
3 • SELECT * FROM employees;
4
```

Result Grid				
			Filter Rows:	Edit:
	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	50000.00
	102	Priya	2	65000.00
	103	Anil	1	55000.00
	104	Sneha	3	70000.00
	105	Kiran	2	48000.00
•	NULL	NULL	NULL	NULL

```

1    DROP VIEW IF EXISTS EmployeeView;
2
3    • CREATE VIEW EmployeeView AS
4    SELECT *
5    FROM employees;
6
7    • SELECT * FROM EmployeeView;
8

```

Result Grid				
		Filter Rows:		Export:  W
	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	50000.00
	102	Priya	2	65000.00
	103	Anil	1	55000.00
	104	Sneha	3	70000.00
	105	Kiran	2	48000.00

```

1 DROP VIEW IF EXISTS EmployeeView;
2
3 • CREATE VIEW EmployeeView AS
4 SELECT *
5 FROM employees;
6
7 • SELECT * FROM EmployeeView;
8

```

Result Grid				
Filter Rows:				
Export: Excel CSV PDF				
	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	50000.00
	102	Priya	2	65000.00
	103	Anil	1	55000.00
	104	Sneha	3	70000.00
	105	Kiran	2	48000.00

```
1 • DROP VIEW IF EXISTS EmployeeBasicView;
2
3 • CREATE VIEW EmployeeBasicView AS
4   SELECT emp_id, emp_name, salary
5   FROM employees;
6
7 • SELECT * FROM EmployeeBasicView;
8
```

Result Grid



Filter Rows:

Export:



	emp_id	emp_name	salary
▶	101	Rahul	50000.00
	102	Priya	65000.00
	103	Anil	55000.00
	104	Sneha	70000.00
	105	Kiran	48000.00

```
1 • DROP VIEW IF EXISTS HighSalaryEmployees;
2
3 CREATE VIEW HighSalaryEmployees AS
4 SELECT emp_id, emp_name, salary
5 FROM employees
6 WHERE salary > 55000;
7
8 • SELECT * FROM HighSalaryEmployees;
9
```

Result Grid			
Filter Rows:			
Export: Export			
Wrap			
	emp_id	emp_name	salary
▶	102	Priya	65000.00
	104	Sneha	70000.00

```

3 • CREATE VIEW EmployeeDepartmentView AS
4 SELECT
5     e.emp_id,
6     e.emp_name,
7     d.dept_name,
8     e.salary
9 FROM employees e
10 JOIN departments d
11 ON e.dept_id = d.dept_id;
12
13 • SELECT * FROM EmployeeDepartmentView;
14

```

Result Grid



Filter Rows:

Export:



Wra

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	102	Priya	ECE	65000.00
	103	Anil	CSE	55000.00
	104	Sneha	EEE	70000.00
	105	Kiran	ECE	48000.00

```
4      SELECT
5          e.emp_id,
6          e.emp_name,
7          d.dept_name,
8          e.salary
9      FROM employees e
10     JOIN departments d
11     ON e.dept_id = d.dept_id
12     WHERE d.dept_name = 'CSE';
13
14 •  SELECT * FROM CSEEmployees;
15
```

Result Grid   Filter Rows:

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	103	Anil	CSE	55000.00

```

4 • CREATE VIEW DepartmentSalaryView AS
5 SELECT
6     d.dept_name,
7     COUNT(e.emp_id) AS EmployeeCount,
8     AVG(e.salary) AS AverageSalary
9 FROM employees e
10 JOIN departments d
11 ON e.dept_id = d.dept_id
12 GROUP BY d.dept_id, d.dept_name;
13
14 • SELECT * FROM DepartmentSalaryView;
15

```

Result Grid



Filter Rows:

Export:



Wrap C

	dept_name	EmployeeCount	AverageSalary
►	CSE	2	52500.000000
	ECE	2	56500.000000
	EEE	1	70000.000000

```

3 • CREATE VIEW HighAverageDepartments AS
4 SELECT
5     d.dept_name,
6     AVG(e.salary) AS AverageSalary
7 FROM employees e
8 JOIN departments d
9 ON e.dept_id = d.dept_id
10 GROUP BY d.dept_id, d.dept_name
11 HAVING AVG(e.salary) > 55000;
12
13 • SELECT * FROM HighAverageDepartments;
14

```

Result Grid |   Filter Rows: | Export:  | Wrap C

dept_name	AverageSalary
ECE	56500.000000
EEE	70000.000000

```
1
2 • DROP VIEW IF EXISTS SalaryRangeView;
3
4 • CREATE VIEW SalaryRangeView AS
5   SELECT emp_id, emp_name, salary
6   FROM employees
7   WHERE salary BETWEEN 50000 AND 70000;
8
9 • SELECT * FROM SalaryRangeView;
10
11
```

Result Grid			
  Filter Rows:			
Export:  Wrap			
	emp_id	emp_name	salary
▶	101	Rahul	50000.00
	102	Priya	65000.00
	103	Anil	55000.00
	104	Sneha	70000.00

```

1 • DROP VIEW IF EXISTS AnnualSalaryView;
2
3 • CREATE VIEW AnnualSalaryView AS
4     SELECT
5         emp_id,
6         emp_name,
7         salary AS MonthlySalary,
8         salary * 12 AS AnnualSalary
9     FROM employees;
10
11 • SELECT * FROM AnnualSalaryView;
12

```

Result Grid |   Filter Rows: | Export: 

	emp_id	emp_name	MonthlySalary	AnnualSalary
▶	101	Rahul	50000.00	600000.00
	102	Priya	65000.00	780000.00
	103	Anil	55000.00	660000.00
	104	Sneha	70000.00	840000.00
	105	Kiran	48000.00	576000.00

```
2
3 • CREATE VIEW EmployeeSalaryView AS
4   SELECT emp_id, emp_name, salary
5   FROM employees;
6
7 • UPDATE EmployeeSalaryView
8   SET salary = 60000
9   WHERE emp_id = 101;
10
11 • SELECT * FROM EmployeeSalaryView;
12 • SELECT * FROM employees WHERE emp_id = 101;
13
```

Result Grid			Filter Rows:	Edit: 		
	emp_id	emp_name	dept_id	salary		
▶	101	Rahul	1	60000.00		
*	NULL	NULL	NULL	NULL		

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WEEK-4 MYSQL STORED PROCEDURES

Setup: employee table

```
1 • CREATE DATABASE IF NOT EXISTS employeeedb1;
2 • USE employeeedb1;
3
4 • CREATE TABLE IF NOT EXISTS employees (
5     emp_id INT PRIMARY KEY,
6     emp_name VARCHAR(50),
7     dept_id INT,
8     salary DECIMAL(10,2)
9 );
10
11 • INSERT INTO employees VALUES
12     (101,'Rahul',1,50000),
13     (102,'Priya',2,65000),
14     (103,'Anil',1,55000),
15     (104,'Sneha',3,70000),
16     (105,'Kiran',2,48000);
17
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	12:41:53	CREATE DATABASE IF NOT EXISTS employeeedb1	1 row(s) affected
✓ 2	12:41:53	USE employeeedb1	0 row(s) affected
✓ 3	12:41:53	CREATE TABLE IF NOT EXISTS employees (emp_id INT PRIMARY KEY, emp_name VARCHAR(50), dept...	0 row(s) affected
✓ 4	12:41:53	INSERT INTO employees VALUES (101,'Rahul',1,50000), (102,'Priya',2,65000), (103,'Anil',1,55000), (104,'Sneha',3,...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

verify data

```
1 • SELECT * FROM employees;
```

Result Grid

Filter Rows:

Edit:

Ex

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	50000.00
	102	Priya	2	65000.00
	103	Anil	1	55000.00
	104	Sneha	3	70000.00
	105	Kiran	2	48000.00
✱	NULL	NULL	NULL	NULL

Display all employees

```
1 DROP PROCEDURE IF EXISTS GetEmployees;
2 DELIMITER $$
3
4 • CREATE PROCEDURE GetEmployees()
5 BEGIN
6     SELECT * FROM employees;
7 END $$
8
9 DELIMITER ;
10
11 • CALL GetEmployees();
12
```

Result Grid					Filter Rows:	Export:	Wrap Cell Content:
	emp_id	emp_name	dept_id	salary			
▶	101	Rahul	1	50000.00			
	102	Priya	2	65000.00			
	103	Anil	1	55000.00			
	104	Sneha	3	70000.00			
	105	Kiran	2	48000.00			

in parametre

```
1 • DROP PROCEDURE IF EXISTS GetEmployeeById;
2 DELIMITER $$
3
4 • CREATE PROCEDURE GetEmployeeById(IN eid INT)
5 BEGIN
6     SELECT * FROM employees
7     WHERE emp_id = eid;
8 END $$
9
10 DELIMITER ;
11
12 • CALL GetEmployeeById(102);
13
14
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	emp_id	emp_name	dept_id	salary
►	102	Priya	2	65000.00

update salary

```
4
5 • CREATE PROCEDURE IncreaseSalary(
6     IN eid INT,
7     IN amount DECIMAL(10,2)
8 )
9 BEGIN
10     UPDATE employees
11     SET salary = salary + amount
12     WHERE emp_id = eid;
13 END $$
14
15 DELIMITER ;
16
17 • CALL IncreaseSalary(101,5000);
18 • SELECT * FROM employees WHERE emp_id=101;
19
```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	55000.00
*	NULL	NULL	NULL	NULL

Filter Rows: Edit:    Export

join

```

2 • DROP PROCEDURE IF EXISTS AvgSalary;
3 DELIMITER $$
4
5 • CREATE PROCEDURE AvgSalary()
6 BEGIN
7     SELECT AVG(salary) AS AverageSalary
8     FROM employees;
9 END $$
10
11 DELIMITER ;
12
13 • CALL AvgSalary();
14

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content
	AverageSalary			
▶	58600.000000			

Group by and avg

```
8
9 • DROP PROCEDURE IF EXISTS EmployeeDepartment;
10 DELIMITER $$
11
12 • CREATE PROCEDURE EmployeeDepartment()
13 BEGIN
14     SELECT e.emp_name, d.dept_name, e.salary
15     FROM employees e
16     JOIN departments d ON e.dept_id=d.dept_id;
17 END $$
18
19 DELIMITER ;
20
21 • CALL EmployeeDepartment();
22
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	emp_name	dept_name	salary
▶	Rahul	CSE	55000.00
	Priya	ECE	65000.00
	Anil	CSE	55000.00
	Sneha	EEE	70000.00
	Kiran	ECE	48000.00

If else :

```
4 • CREATE PROCEDURE DepartmentAverageSalary()
5 BEGIN
6     SELECT d.dept_name,
7           COUNT(e.emp_id) AS EmployeeCount,
8           AVG(e.salary) AS AverageSalary
9     FROM employees e
10    JOIN departments d ON e.dept_id=d.dept_id
11   GROUP BY d.dept_id, d.dept_name;
12 END $$
13
14 DELIMITER ;
15
16 • CALL DepartmentAverageSalary();
17
```

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
	dept_name	EmployeeCount	AverageSalary
▶	CSE	2	55000.000000
	ECE	2	56500.000000
	EEE	1	70000.000000

```

8      SELECT salary INTO sal
9      FROM employees
10     WHERE emp_id=eid;
11
12     IF sal >= 60000 THEN
13         SELECT 'High Salary' AS Status;
14     ELSE
15         SELECT 'Low Salary' AS Status;
16     END IF;
17 END $$
18
19 DELIMITER ;
20
21 • CALL SalaryStatus(102);
22

```

Result Grid		Filter Rows:	Export:	Wrap Cell Co
	Status			
▶	High Salary			

Out parameter

Inout parameter

```
1 • DROP PROCEDURE IF EXISTS EmployeeCount;
2 DELIMITER $$
3
4 • CREATE PROCEDURE EmployeeCount(OUT total INT)
5 BEGIN
6     SELECT COUNT(*) INTO total
7     FROM employees;
8 END $$
9
10 DELIMITER ;
11
12 • CALL EmployeeCount(@total);
13 • SELECT @total AS TotalEmployees;
14
15
```

Result Grid		Filter Rows:	Export:	Wrap Cell Cont
	TotalEmployees			
▶	5			

transaction

```
1 • DROP PROCEDURE IF EXISTS AddBonus;
2 DELIMITER $$
3
4 • CREATE PROCEDURE AddBonus(INOUT bonus DECIMAL(10,2))
5 BEGIN
6     SET bonus = bonus + 5000;
7 END $$
8
9 DELIMITER ;
10
11 • SET @bonus=10000;
12 • CALL AddBonus(@bonus);
13 • SELECT @bonus AS UpdatedBonus;
14
15
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	UpdatedBonus
▶	15000.00

```

13      SET salary=salary-amount
14      WHERE emp_id=fromEmp;
15
16      UPDATE employees
17      SET salary=salary+amount
18      WHERE emp_id=toEmp;
19
20      COMMIT;
21  END $$
22
23  DELIMITER ;
24
25 • CALL TransferSalaryAmount(101,102,2000);
26 • SELECT * FROM employees WHERE emp_id IN (101,102);
27

```

Result Grid				
Filter Rows:				
Edit:    Export/Import:  				
	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	53000.00
	102	Priya	2	67000.00
✱	NULL	NULL	NULL	NULL

```

1 • SHOW PROCEDURE STATUS
2   WHERE Db='employeeeb1';
3
4   SHOW CREATE PROCEDURE GetEmployees;
5
6 • DROP PROCEDURE IF EXISTS GetEmployees;
7

```

Result Grid					
Filter Rows:					
Export:  Wrap Cell Content: 					
	Procedure	sql_mode	Create Procedure	character_set_client	collation_connection
▶	GetEmployees	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	CREATE DEFINER='root'@'localhost' PROCE...	utf8mb4	utf8mb4_0900_ai_ci
					utf8mb4_0900_ai_ci

